MAHMOUD ELBALAT

AI ENGINEER | MACHINE LEARNING | COMPUTER VISION

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Portfolio: https://mahmoudsalah2002.github.io/Portfolio/

EDUCATION

Bachelor's Degree in Computer Engineering | *Menoufia University, Egypt | July 2025*
Graduation grade: Good

LANGUAGES

Arabic: Fluent **English:** Intermediate (B1)

PROFESSIONAL SUMMARY

Computer Engineering graduate with hands-on experience in machine learning and AI projects, including model training and evaluation, predictive modeling, NLP, and computer vision. Practical experience with YOLOv8 object detection, image annotation, and OpenCV-based inference, with an interest in building and deploying reliable AI solutions.

TECHNICAL SKILLS

Programming | *Python, C++, Java, SQL, Assembly*

AI / Machine Learning | *Machine Learning, Deep Learning, model training and evaluation, predictive modeling, NLP*

Computer Vision | *YOLOv8, object detection, image annotation, OpenCV*

Frameworks & Libraries | *PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Matplotlib, Flask*

Tools | *CVAT, Git, GitHub*

PROJECTS

YOLOv8 Cat & Dog Object Detection | *Computer Vision*

- Prepared and annotated a cat-and-dog dataset in YOLO format and organized it into training, validation, and test splits.
- Trained and evaluated a YOLOv8 object-detection model and reviewed evaluation outputs, including confusion matrices and precision–recall results.
- Tested predictions on sample images and webcam input.
- Published the project code, usage instructions, example detections, and trained model weights in a GitHub repository.

GitHub: https://github.com/MahmoudSalah2002/YOLOv8-Object-Detection

Candy Detection with YOLOv8 | *Computer Vision*

- Worked on an object-detection project using YOLOv8 to identify candy packages across five classes: BabyRuth, Butterfinger, CRUNCH, MilkyWay, and TWIX.
- Labeled images and used the dataset to support object-detection training and testing.

Smart Irrigation System | *Graduation Project*

- Developed a machine-learning workflow to predict irrigation needs using temperature, soil moisture, and soil temperature features.
- Compared Decision Tree, Linear Regression, and Neural Network approaches; selected Linear Regression based on the project's evaluation.
- Designed the system to support irrigation decisions and improve water-use efficiency.

Conversational Chatbot | *NLP*

- Developed a Python-based chatbot prototype using NLP techniques to support automated customer interactions.
- Worked with intent recognition and entity extraction and explored integration with a company website.
- The original project notes reported a 40% improvement in response time and a 25% reduction in manual queries.

EXPERIENCE & TRAINING

Machine Learning Intern | *Poornima University, India | July 2024 – September 2024*

- Developed a Random Forest model for customer-churn prediction; the original CV reports 85% accuracy.
- Developed, trained, and evaluated machine-learning models and analyzed datasets to identify patterns and improve model performance.
- Gained exposure to reinforcement-learning algorithms through work with different models.

AI & Machine Learning Diploma | *AMIT Learning | June 2023 – August 2023*

- Studied Python programming, data science, machine learning, deep learning, computer vision, and NLP.
- Completed practical applications and projects involving datasets and AI/ML techniques; explored MLOps concepts.

LEADERSHIP & ACTIVITIES

Organizing Committee Member, IEEE | *October 2023 – Present*

- Helped organize and coordinate events, workshops, and community initiatives.

Global Volunteer, AIESEC | *March 2024 – November 2024*

- Participated in an international volunteer project in India and collaborated with people from diverse backgrounds.

CERTIFICATIONS

- Machine Learning Internship Certificate — Poornima University (2024)
- AI & Machine Learning — AMIT Learning (certificate year listed in original CV: 2025)